MotionScoring-VLM: Reading Out Trajectory Quality from a Frozen Driving VLM as Constrained Filtering

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Abstract

Driving evaluators increasingly use vision-language models (VLMs) to recognize scene-dependent reasons that fixed rule tables miss, but existing systems usually obtain scoring ability by modifying the backbone through LoRA, supervised finetuning, reinforcement learning, or full-stack adaptation. We study the complementary hypothesis that a strong driving VLM may already encode substantial trajectory-quality information and primarily lack a reliable native readout. The architecture suggests a precise formulation: the attention branch of a trailing score query is a content-dependent weighted aggregation of preceding frozen context states. Under an explicit signal–interference model, this aggregation has the same weighted structure as classical filter design. The transfer is not direct, however, because classical taps are free while Transformer attention is restricted to patterns generated by shared trainable query embeddings through a frozen network. We define the resulting per-instance realizable filter set $\mathcal{A}_\Phi(x,\tau) = \{\alpha_\Phi(E;x,\tau) : E \in \mathcal{E}\}$, with one shared E optimized over the scene distribution. MotionScoring-VLM uses an SLQ-style query bank to parameterize this constrained space, a scalar score head trained from automatically generated DriveJudge pseudo labels, and a teacher-conditioned deflection objective that rewards pseudo-score-aligned variation in the attention-filtered latent while whitening teacher-orthogonal residual variation. The method changes no backbone weight and requires no human label, preference pipeline, anchoring set, assessment-generation stage, ordinal reporting module, or reference-cancellation branch. We pre-register the comparisons that separate the claims: direct prompting versus SLQ tests the readout thesis; SLQ versus SLQ plus filtering tests whether filter theory contributes algorithmically; and LoRA tests whether scoring ability must be installed into weights. Human-labeled data are used only for independent evaluation, including a secondary weak-to-strong stress test on teacher–human disagreements.

1 Problem setting: trajectory quality may be represented but not natively readable

Let x collect the multimodal driving context (camera or rendered scene, command, and ego state), and let τ be a candidate trajectory represented in the VLM input. A frozen driving VLM Φ maps the sequence to contextual states

$$H(x,\tau) = \Phi(x,\tau) = (h_1,\dots,h_n), \qquad \Phi \text{ fixed.} \tag{1}$$

A conventional learned evaluator changes Φ so that a scoring head can predict a trajectory score. Our thesis asks whether this is always necessary. The empirical starting point consists of two pre-registered measurements on the same checkpoint:

1. a lightweight probe on frozen hidden states should recover substantially more human-aligned trajectory ordering than a rule score or direct score prompt; and
2. direct scalar prompting should remain substantially below that probe ceiling, even under prompt and decoding controls.

If the probe is weak, the representation premise fails and the paper should not be framed as readout recovery. If the probe is strong but direct prompting is weak, the gap isolates a deployable question that ordinary probing does not answer: *which quality information can the model access through its own frozen causal readout interface?*

This distinction also separates our problem from standard representation probing. A probe permits an arbitrary external predictor class and asks whether information is decodable. MotionScoring instead fixes the model and studies the readouts realizable through appended native queries. The former estimates a representation ceiling; the latter is a constrained-interface problem.

The supervision source is a rule-gated teacher, DriveJudge, which automatically maps a candidate to a pseudo score

$$y^*(x,\tau) = T_{\mathrm{DJ}}(x,\tau) \in [0,1]. \tag{2}$$

No human annotation enters training. The teacher supplies a scalable target, not a claim of human optimality; its known rule bottleneck is handled as a separate evaluation question in Section 6.

2 From a trailing query to a constrained latent filter

2.1 The exact architectural object: an attention-weighted context readout

Append K trainable query embeddings and designate the final one as the score query,

$$E = (e_1,\dots,e_K,e_s) \in \mathbb{R}^{(K+1)\times d}. \tag{3}$$

Because these positions trail the original context in a causal model, the original context states are unchanged by the append operation, while the queries can attend backward to them. For one final-layer head r, the *context contribution* to the score-query attention branch is exactly

$$o_E^{(r)}(x,\tau) = \sum_{i \in C(x,\tau)} \alpha_{i,\Phi}^{(r)}(E; x,\tau)\, v_i^{(r)}(x,\tau), \quad \alpha_\Phi^{(r)} \in \Delta^{|C|-1}, \tag{4}$$

where C indexes original multimodal context positions and $v_i^{(r)}$ are their frozen value projections. Combining heads through the frozen output projection gives

$$z_E(x,\tau) = \sum_{r=1}^{R_h} W_O^{(r)} o_E^{(r)}(x,\tau). \tag{5}$$

Equation (4), not the entire Transformer state, is the exact filtering object: a content-dependent weighted aggregation over frozen context values. Residual paths, MLPs, and query-to-query computation remain part of the scoring network, but the regularizer below is attached specifically to z_E, where the filter interpretation is literal.

2.2 A conditional signal–interference model

Along any score-relevant linear direction w, model the token contributions as

$$w^\top W_O^{(r)} v_i^{(r)} = \theta_i + \epsilon_i, \tag{6}$$

where θ_i is the trajectory-quality component and ϵ_i collects irrelevant semantics, formatting effects, and other nuisance variation. This is the paper's sole modeling assumption; the method must be ablated if the resulting filter statistic does not track held-out scoring performance.

Under (6), the readout contains

$$w^\top z_E = \alpha_E^\top \theta + \alpha_E^\top \epsilon. \tag{7}$$

If $\Sigma = \mathrm{Cov}(\epsilon)$ and the taps were freely designable, the classical output deflection ratio would be

$$\rho(\alpha) = \frac{(\alpha^\top \theta)^2}{\alpha^\top \Sigma \alpha}, \qquad \alpha^*_{\text{free}} \propto \Sigma^{-1}\theta, \tag{8}$$

up to the constraints imposed on α. Equation (8) is a reference optimum, not our algorithm: neither θ nor Σ is observed, and a frozen Transformer does not permit free tap assignment.

2.3 The true optimization space

For a fixed scene and trajectory, varying the shared query embeddings produces only

$$\mathcal{A}_\Phi(x,\tau) = \{\alpha_\Phi(E; x,\tau) : E \in \mathcal{E}\} \subseteq \prod_{r=1}^{R_h} \Delta^{|C|-1}. \tag{9}$$

The classical optimum in (8) is therefore not guaranteed to be realizable. More importantly, E is not re-optimized per example. MotionScoring learns one shared readout rule:

$$E^* = \arg\min_{E \in \mathcal{E}} \mathbb{E}_{(x,\tau)\sim\mathcal{D}}\big[\mathcal{L}(x,\tau; E)\big]. \tag{10}$$

Proposition 1 (Frozen-model realizability constraint). With Φ fixed, every trainable score-query context-attention pattern lies in (9), and optimizing shared E in (10) learns a distribution-level readout rather than per-instance taps. Any unconstrained filter optimum serves only as an upper/reference solution unless it belongs to $\mathcal{A}_\Phi(x,\tau)$ for the relevant inputs.

The proposition is definition-level but consequential: it transforms "apply a matched filter" into a new problem, *learn the best filter realizable through the native attention geometry of a frozen VLM*. SLQ supplies a validated parameterization of such frozen native readouts; our contribution is the trajectory-scoring objective that makes this parameterization a constrained semantic filter rather than a retrieval embedding.

3 MotionScoring-VLM

3.1 SLQ parameterization and scalar scoring

The frozen model processes (x,τ) once, followed by the appended query bank. Let $h_s(E; x,\tau)$ denote the final score-query state. A lightweight scalar head g_ω predicts

$$\widehat{y}_E(x,\tau) = g_\omega(h_s(E; x,\tau)). \tag{11}$$

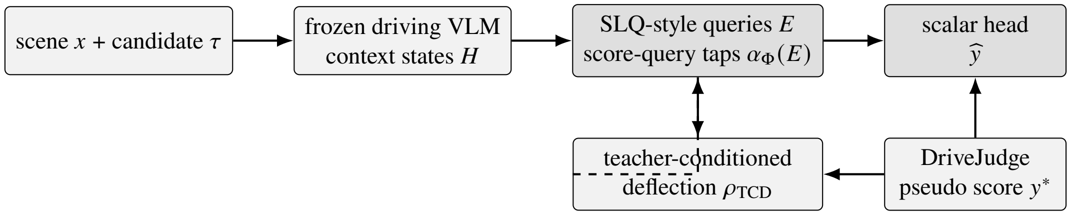

Figure 1: MotionScoring-VLM. The VLM is frozen. Pseudo-score loss trains what to predict; the teacher-conditioned deflection term acts on the attention-filtered context latent and selects how the shared native readout should extract it.

Only E and ω are trainable. The exact parameter count is determined by K, hidden width, and head size; the intended setting is sub-million and is reported rather than assumed. The basic pseudo-label objective is

$$\mathcal{L}_{\text{score}}(E, \omega) = \frac{1}{B} \sum_{b=1}^{B} \ell\big(\widehat{y}_b, y_b^*\big), \tag{12}$$

with Huber or squared loss on the DriveJudge scale.

Plain SLQ scoring would stop at (12). It can change attention because gradients propagate through $h_s \to \alpha_\Phi(E) \to E$, but a scalar target does not directly identify which internal readout should realize that target. MotionScoring adds a filter-derived selection principle on z_E.

3.2 Teacher-conditioned deflection: a computable filter objective

For a minibatch, let $z_b = z_E(x_b, \tau_b) \in \mathbb{R}^d$ be the exact context-attention branch output in (5), and center the pseudo scores and filtered latents:

$$\widetilde{y}_b = y_b^* - \overline{y}^*, \qquad \widetilde{z}_b = z_b - \overline{z}. \tag{13}$$

The least-squares teacher-aligned signal slope in the filtered latent is

$$\widehat{s}_E = \frac{\sum_{b=1}^{B} \widetilde{y}_b \widetilde{z}_b}{\sum_{b=1}^{B} \widetilde{y}_b^2 + \varepsilon} \in \mathbb{R}^d. \tag{14}$$

Remove this one-dimensional teacher-aligned component to obtain residuals

$$r_b(E) = \widetilde{z}_b - \widehat{s}_E \widetilde{y}_b, \tag{15}$$

and estimate a stable diagonal nuisance covariance

$$\widehat{D}_E = \operatorname{diag}\left(\frac{1}{B} \sum_{b=1}^{B} r_b(E) \odot r_b(E)\right) + \varepsilon I. \tag{16}$$

For any latent scoring direction w, the teacher-conditioned deflection is

$$\mathcal{R}(w; E) = \frac{(w^\top \widehat{s}_E)^2}{w^\top \widehat{D}_E w}. \tag{17}$$

The maximizer is the familiar whitened direction $w^* \propto \widehat{D}_E^{-1} \widehat{s}_E$, and the maximum is available in closed form:

$$\rho_{\text{TCD}}(E; \mathcal{B}) = \max_{w \neq 0} \mathcal{R}(w; E) = \widehat{s}_E^\top \widehat{D}_E^{-1} \widehat{s}_E. \tag{18}$$

Equation (18) is the practical bridge from pseudo labels to filter design. The pseudo scores estimate which minibatch variation is task signal; the residual covariance estimates variation not explained by that score; and maximizing (18) changes E, hence $\alpha_\Phi(E)$, so that the native context filter exposes a stronger, more stable teacher-aligned latent direction. The diagonal approximation is an explicit computational choice, not a theorem; shrinkage/full-covariance and label-bin Fisher variants are ablations.

The joint objective is

$$(E^*, \omega^*) = \arg\min_{E, \omega} \left[\mathcal{L}_{\text{score}}(E, \omega) - \lambda\, \rho_{\text{TCD}}(E; \mathcal{B})\right]. \tag{19}$$

Because both numerator and denominator of (17) scale quadratically with z_E, the deflection term is approximately scale-invariant and does not simply reward larger hidden norms. The small ridge ε prevents numerical singularities. In implementation, batch moments may be replaced by exponential moving averages; this is a stability choice and is evaluated separately.

3.3 Optimization loop

Algorithm 1: MotionScoring-VLM training

Require: frozen VLM Φ; shared query embeddings E; scalar head g_ω; offline DriveJudge scores y^*; weight λ.
1. Sample a minibatch $\mathcal{B} = \{(x_b, \tau_b, y_b^*)\}_{b=1}^B$.
2. Run the frozen context and append E; extract score state $h_{s,b}$ and context-attention output z_b.
3. Predict $\widehat{y}_b = g_\omega(h_{s,b})$ and compute $\mathcal{L}_{\text{score}}$ in (12).
4. Estimate $\widehat{s}_E$, $\widehat{D}_E$, and ρ_{TCD} using (14)–(18).
5. Update only (E, ω) with the gradient of (19); never update Φ.

Return: one shared readout E^* used for every test scene; no per-instance optimization.

3.4 Implementation and computational cost

DriveJudge scores are generated once and stored, so no teacher call is needed during student optimization. The original multimodal context is frozen and causally precedes the readout queries; its key/value states may therefore be cached. Training adds only the incremental query computation and gradients with respect to (E, ω). For hidden width d, query count $K + 1$, and a linear scalar head, the trainable count is approximately $(K + 1)d + d + 1$; a two-layer head adds only its small bottleneck. The added attention cost per layer is $O((K + 1)n + (K + 1)^2)$ rather than recomputing or differentiating all context-context attention.

For multi-head models, z_E in (5) is obtained from the actual final-layer context-attention branch before the residual addition. The score head still reads the full final score-query state. This separation matters: the task loss may exploit the whole native computation, while the filter term is attached only where the weighted-filter identity is exact. At inference, the filter statistic is not required; the deployed evaluator is simply the frozen VLM, the learned query bank, and the scalar head.

MotionScoring in one sentence. Keep the driving VLM fixed, use SLQ queries to search its realizable native attention-filter space, and use automatic pseudo scores both to train the scalar output and to estimate a deflection criterion that favors a high-signal, low-nuisance latent readout.

4 Why filtering is an algorithmic contribution rather than an analogy

The filtering view is useful only if it changes the learned method beyond plain pseudo-label distillation. The two losses in (19) have different objects:

- $\mathcal{L}_{\text{score}}$ constrains the endpoint $\widehat{y}$ and can be minimized by any readout/head pair that reproduces the teacher scores;
- ρ_{TCD} acts on the internal context-attention output before the scalar head and rewards a readout whose teacher-aligned variation survives nuisance whitening.

Consequently, the filter term is a selection principle over output-compatible SLQ solutions. It does not guarantee human alignment, nor does it make attention uniquely identifiable. Its falsifiable prediction is narrower: at matched teacher-score fit, adding (18) should increase held-out teacher-conditioned deflection and improve generalization, especially under distribution shift where shortcut readouts are less stable.

This positioning clarifies the relation to prior work. Frozen probes show that information can be decoded, but use an external classifier. SLQ shows that appended native queries can elicit useful frozen representations, but is developed for retrieval. MotionScoring transfers the query mechanism to trajectory scoring and introduces a filter-derived objective tied to continuous pseudo supervision. LoRA and full finetuning remain the appropriate alternatives when the frozen representation or its realizable native readout is insufficient.

5 Evaluation: separate representation, readout, filtering, and teacher inheritance

The empirical program is organized so that each comparison answers one claim rather than only populating a leaderboard.

5.1 Tasks, supervision, and metrics

Use the same frozen driving checkpoint for all readout variants. Evaluate: (i) human-aligned pairwise trajectory judgment on a public preference split such as DriveCritic; (ii) pseudo-score fit and ranking on a NAVSIM/Pseudo-Simulation-style candidate set; and (iii) long-tail/OOD regimes where semantic necessity matters. Exact evaluation items are held out from pseudo-label training.

Report pairwise accuracy, Spearman correlation, MAE, and calibration on the teacher scale; task-level trajectory selection when candidates are available; trainable parameters and latency; and the filter diagnostics ρ_{TCD}, attention entropy, and sensitivity to attention interventions. Public human judgments are used only for evaluation.

5.2 Pre-registered verdicts

Readout verdict. If SLQ-Score does not beat direct prompting, the native realizable space may be too restrictive or the optimization may fail. If the frozen probe is also weak, the representation premise is rejected.

Filtering verdict. If MotionScoring does not improve over matched SLQ-Score, or if ρ_{TCD} rises without any held-out benefit,

Method	Backbone update	Query readout	Pseudo labels	Filter objective	Human/OOD metric	Meaning
Direct prompting	no	no	no	no	□	native channel
Frozen linear probe	no	external	yes	no	□	decodability ceiling
SLQ-Score	no	yes	yes	no	□	readout hypothesis
MotionScoring	no	yes	yes	yes	□	filtering contribution
LoRA scorer	yes	–	yes	no	□	weight-installation alternative

Table 1: Pre-registered main comparison. The decisive contrast is SLQ-Score versus MotionScoring under the same frozen checkpoint, pseudo labels, query count, head, and training budget.

constrained filtering remains an interpretive lens rather than an algorithmic contribution. The central positive pattern is

$$\text{Prompting} < \text{SLQ-Score} < \text{MotionScoring}, \tag{20}$$

with the second inequality reproduced across seeds and strongest under shift.

Weight-installation verdict. If MotionScoring approaches a matched LoRA scorer while preserving all backbone behavior by construction, the result supports "read out rather than install." A persistent gap to LoRA quantifies how much scoring ability is not natively accessible.

5.3 Mechanism controls

The following tests prevent attention visualizations from being treated as evidence by themselves:

- **Attention flattening:** replace the learned score-query context attention by a uniform distribution at inference while holding the scalar head fixed. A real filter mechanism predicts a material drop.
- **Query freeze:** train only the scalar head on fixed/random queries. Improvement from training E isolates native attention adaptation from head capacity.
- **Label-permutation control:** compute the filter term from permuted pseudo scores. It should not improve and may reduce generalization; otherwise (18) is acting as generic variance shaping rather than teacher-conditioned filtering.
- **Objective trace:** report ρ_{TCD}, teacher-score fit, and human/OOD metrics together. A useful mechanism predicts that deflection gains accompany, rather than replace, external performance gains.
- **Capacity sweep:** vary K only as an ablation. No theorem claims monotonic expansion; the empirical prediction is improvement followed by saturation if additional query capacity is useful.

6 Secondary stress test: weak-to-strong generalization

DriveJudge is a structurally weaker supervisor in regimes where fixed rules are incomplete. Training on its pseudo scores can therefore transmit its bias; a frozen backbone and a small readout provide no guarantee against inheritance. The only plausible route beyond the teacher is that its labels underdetermine which frozen semantic features realize the score, while the deflection objective favors a stable teacher-aligned latent direction that may generalize differently outside the teacher's reliable regime.

We test this as a secondary consequence, not a premise of the core method. On a public human-labeled benchmark, isolate held-out items where DriveJudge and humans disagree and compare the student to both. Exact evaluation items never enter training, although the same scenario types may. Define

$$\text{TranscendRate} = \frac{1}{|\mathcal{D}_{\text{dis}}|} \sum_{i \in \mathcal{D}_{\text{dis}}} \mathbf{1}\left[|\widehat{y}_i - y_i^{\text{human}}| < |y_i^{*} - y_i^{\text{human}}| \right]. \tag{21}$$

A rate above appropriate calibration and plain-SLQ baselines supports weak-to-strong generalization. Full inheritance does not invalidate constrained filtering; it limits the paper to honest pseudo-label distillation. No human label is used for training or model selection in this test.

7 Claims, scope, and conclusion

Scope. The method is capped twice: first by what trajectory-quality information the frozen representation contains, and second by what its native query interface can realize. A trained external probe estimates the first ceiling; the gap from MotionScoring to that probe estimates remaining readout shortfall. The method optimizes one simple corner of native readout design: shared SLQ query embeddings, a scalar head, and a diagonal teacher-conditioned deflection criterion. It does not claim that attention is the unique explanation of a score, that teacher supervision guarantees human validity, or that filtering always improves over ordinary distillation.

Claim level	Statement	Kill switch / required evidence
Exact architecture	Eq. (4) is a weighted context aggregation; all trainable taps are generated by shared E through frozen Φ.	Implementation audit; causal masking must leave original context states unchanged.
Conditional model	The readout can be studied as quality signal plus nuisance, yielding the filter criterion in (8).	ρ_{TCD} must track held-out behavior and survive label-permutation controls.
Method claim	The teacher-conditioned deflection objective improves plain SLQ under matched supervision and capacity.	SLQ versus MotionScoring across seeds and shifts.
Empirical premise	Frozen states contain substantial trajectory-quality information.	Probe ceiling must exceed prompting and rule scores.
Secondary hypothesis	The student may outperform the weak teacher on human disagreement items.	Eq. (21); failure downgrades, not invalidates, the core paper.

Table 2: Claim registry. Exact, conditional, algorithmic, and empirical statements are deliberately separated.

Takeaway. The paper begins from a driving-scoring failure, derives the weighted aggregation implemented by a trailing query, recognizes its conditional signal–interference structure, and then discovers the constraint that makes the problem distinct from classical filtering: the taps must lie in $\mathcal{A}_\Phi$ and be generated by one shared readout across all scenes. SLQ provides the optimization substrate; pseudo labels reveal the otherwise unobserved quality direction; and the teacher-conditioned deflection objective selects a high-signal, low-nuisance native readout. The core lesson is concise:

$$\text{quality information in frozen states} \quad \not\Rightarrow \quad \text{quality accessible through the native score channel.} \tag{22}$$

MotionScoring-VLM turns that gap into a concrete constrained-filtering problem and a minimal, fully automatic training algorithm.

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